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P2 — Sustainable Energy

OCR GCSE Physics B (J259) — 21st Century Science · Paper 1 (Breadth in Physics)

P2.1 Energy Stores and Transfers

Energy cannot be created or destroyed — it is only **transferred** between stores (the **Principle of Conservation of Energy**). What we call "using energy" really means *transferring* energy from a useful store to less useful ones (often into the thermal store of the surroundings).

The Eight Energy Stores

Store	Example
Kinetic	Anything moving — a running athlete, a flying cricket ball
Gravitational potential	An object raised in a gravitational field — a book on a shelf
Elastic potential	A stretched spring, a drawn bow, a rubber band
Thermal (internal)	A hot cup of tea, the Sun
Chemical	Food, fuels, batteries
Magnetic	Two repelling magnets held apart
Electrostatic	Charged Van de Graaff, thunderclouds
Nuclear	Stored in the nuclei of atoms (fission/fusion fuels)

The Four Pathways

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Energy is moved *between* stores by four pathways:

- **Mechanical work** — a force doing work (pushing, lifting, stretching).
- **Electrical work** — charges being moved by a p.d.
- **Heating** — energy transferred from hot to cold (particle collisions, radiation).
- **Radiation** — energy carried by waves (light, sound, IR).

P2.2 Calculating Energy Changes

Kinetic energy: $E_k = \frac{1}{2} m v^2$

$(J) = (kg) \times (m/s)^2$

Gravitational potential energy: $E_p = m g h$

$(J) = (kg) \times (N/kg) \times (m)$ $(g \approx 9.8\ N/kg$ on Earth$)$

Elastic potential energy: $E_e = \frac{1}{2} k x^2$

$(J) = (N/m) \times (m)^2$

Heating (change in thermal store): $\Delta E = m c \Delta\theta$

$(J) = (kg) \times (J/kg\ °C) \times (°C)$

Worked Example: A 70 kg cyclist is moving at 8 m/s. How much kinetic energy do they have?

$E_k = \frac{1}{2} \times 70 \times 8^2 = \frac{1}{2} \times 70 \times 64 =$ **2240 J**

Worked Example: A 2 kg book is lifted 1.5 m onto a shelf. How much GPE does it gain?

$E_p = m g h = 2 \times 9.8 \times 1.5 =$ **29.4 J**

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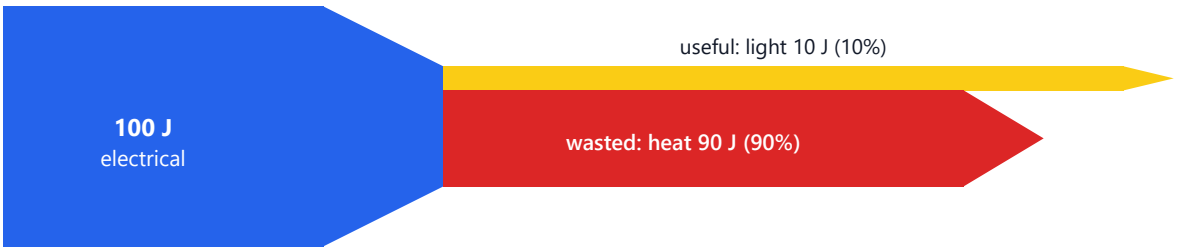
P2.3 Efficiency and Sankey Diagrams

The **efficiency** of a device is the fraction (or %) of the input energy that is transferred **usefully**. The rest is *wasted* — usually to the thermal store of the surroundings.

Efficiency = useful energy output / total energy input
= useful power output / total power input

Sankey Diagram — Filament Lamp (100 J input)



Efficiency = 10/100 = 0.10 = 10% — that's why LEDs (≈ 40–90%) have replaced filament bulbs.

Worked Example: A motor lifts a 5 kg load 2 m. If the motor uses 200 J of electricity, what is its efficiency?
Useful output = m g h = 5 × 9.8 × 2 = 98 J. Efficiency = 98/200 = **0.49 (49%)**

P2.4 Power

Power is the rate at which energy is transferred (or the rate at which work is done). Unit: **watts (W)**, where 1 W = 1 J/s.

$$P = E / t \quad \text{or} \quad P = W / t$$

$$(W) = (J) / (s)$$

Worked Example: A hairdryer transfers 36 000 J in 30 s. What is its power?
P = E / t = 36 000 / 30 = **1200 W (1.2 kW)**

P2.5 Reducing Unwanted Energy Transfers

In our homes, most wasted energy leaves through the walls, roof, windows and floors as heat. We can reduce this with:

Method	How it works
Loft insulation (fibreglass)	Traps air — a poor conductor — reducing conduction through the ceiling
Cavity wall insulation (foam)	Reduces conduction and convection currents inside the wall cavity
Double glazing	The gas/vacuum gap between the panes is a very poor conductor
Draught excluders	Prevent heat loss by air exchange (convection)
Thicker walls / lower thermal conductivity	Reduce the rate of heat conduction

Rate of cooling depends on: thickness, thermal conductivity, surface area, and temperature difference between inside and outside.

Exam Tip: Questions often ask about *payback time*: time = cost of insulation / annual saving. A £200 loft insulation that saves £50 a year pays back in 4 years.

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P2.6 Energy Resources

An **energy resource** is a naturally occurring store from which useful energy can be extracted — usually to make electricity, heat buildings or move vehicles.

Renewable vs Non-Renewable

- **Non-renewable**: will run out because they are used faster than they form. Fossil fuels (coal, oil, gas) and nuclear fuel (uranium).
- **Renewable**: replenished naturally faster than they are used. Wind, solar, hydroelectric, tidal, wave, geothermal, biofuels.

Resource	How it's used	Advantages	Disadvantages
Coal / oil / gas	Burnt to boil water → steam turns turbine → generator	Reliable, high energy density, established infrastructure	CO_2 → climate change; SO_2 → acid rain; non-renewable
Nuclear (fission of uranium)	Heat from fission boils water → turbine → generator	No CO_2; very large energy per kg of fuel; reliable	Radioactive waste (very long-lived); high decommissioning cost; accident risk
Wind	Wind turns turbine blades → generator	No fuel, no CO_2, free	Intermittent (no wind = no power); visual/noise impact on landscape
Solar (PV)	Photovoltaic cells convert sunlight directly to electricity	No fuel, no CO_2, modular	Only works in daylight; efficiency drops on cloudy days
Hydroelectric	Water stored behind a dam turns a turbine as it falls	Reliable, can ramp up quickly, no fuel	Drowns valleys (habitat loss), methane from rotting vegetation, very expensive

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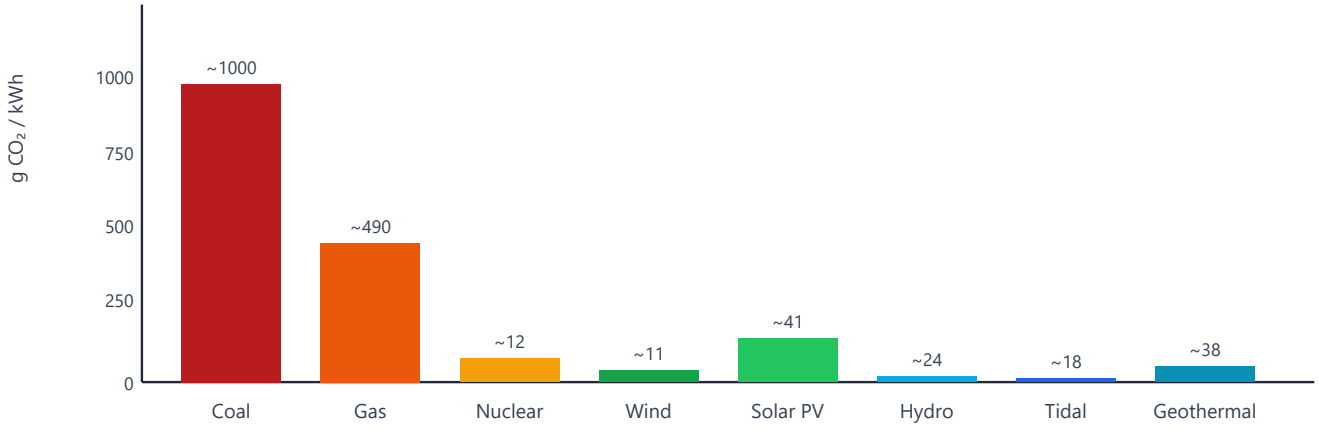
Tidal	Tidal movement turns turbines (e.g. estuary barrage)	Predictable, no fuel, no CO_2 at point of use	Only certain coastal locations; disrupts estuary ecology
Wave	Rising/falling water drives generators	Renewable, no fuel	Hazard to shipping; variable output
Geothermal	Uses heat from hot rocks to boil water	Reliable, small footprint, low CO_2	Few suitable locations (Iceland, NZ, parts of Italy)
Biofuel (ethanol, biodiesel)	Burn plant/animal matter or ferment to alcohol	Roughly carbon-neutral if crop is re-grown	Competes with food crops for land; deforestation in some cases

Comparing Resources

Typical CO_2 emissions per kWh electricity generated



(Lifecycle CO_2, source: IPCC — indicative values)

P2.7 Trends in Electricity Generation

The UK electricity mix has shifted dramatically since 1990:

- Coal: fell from > 70% of generation to almost zero by 2024.
- Gas: rose through the 1990s; still the largest single source.

- Renewables (wind + solar + biomass + hydro): rose from ~2% to ~40%+.
- Nuclear: steady, around 15%.

Exam Tip: A standard 6-mark question: "*Compare the environmental and economic advantages/disadvantages of two named energy resources.*" Always name **specific** drawbacks (CO_2, acid rain, habitat loss, radioactive waste) — not generic "bad for environment".

P2.8 Power Stations & the National Grid (overview)

Thermal Power Station Energy Flow

Waste heat is lost to the surroundings (often via cooling towers) — this limits efficiency to ≈ 35–45%.

Exam Tip: Wind, hydro, tidal and wave skip the "heat → steam" step — they drive turbines directly, giving higher overall efficiency.